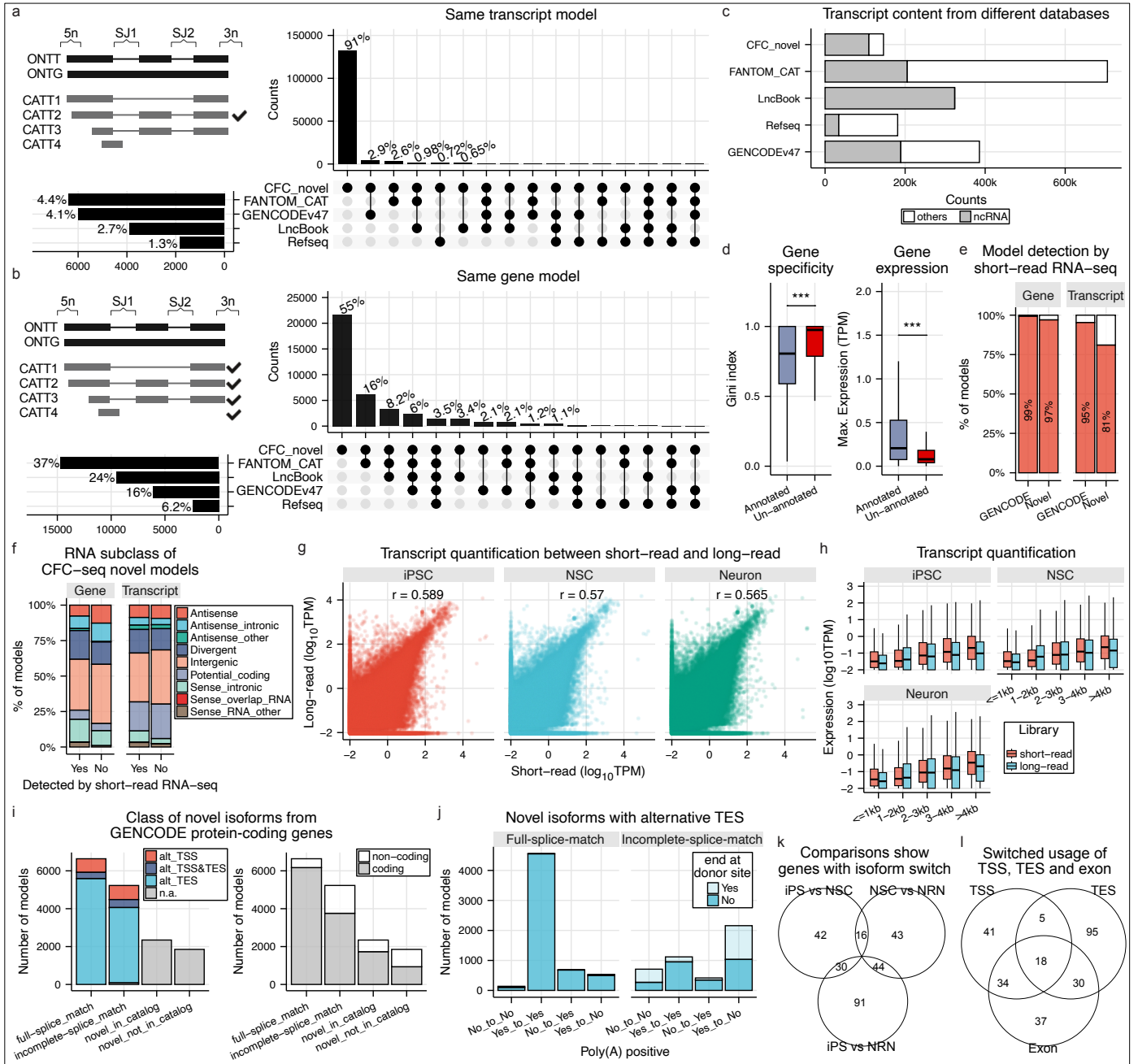




Extended Data Figure 5 | Transcript and gene models identified by SALA.

a, UpSet plot showing novel transcript models identified in this study and their overlap with FANTOM CAT, LncBook, RefSeq, and GENCODE v47. Transcript model identity was determined by SALA; schematic shown on the left. **b**, UpSet plot of novel gene models and their overlap with the same four databases, as identified by SALA. **c**, Counts of transcript models annotated as ncRNA or other classes across the four databases. **d**, Cell-type specificity (Gini index) of gene models annotated in any of the public databases or completely novel, across the five studied cell types. Right, maximum expression level across the five cell types for each group. **e**, Proportion of gene and transcript models in the final CFC-seq transcriptome also detected by short-read RNA-seq in iPSC, NSC, and Neuron. **f**, Classification of novel gene and transcript models (detected or undetected by short-read RNA-seq) into ncRNA subclasses. **g**, Correlation of expression levels from long- and short-read data using the finalised transcriptome, limited to detectable GENCODE models. **h**, Expression levels of transcript models from short- and long-read libraries, grouped by transcript length. **i**, Classification of novel isoforms of protein-coding genes. "Full-splice match" indicates isoforms with all splice junctions matching a GENCODE transcript; "incomplete-splice match" includes partial matches. Alternative TSS and TES was assessed by ex5_clusters and ex3_clusters relative to best-matched GENCODE transcripts; skipped for unmatched cases (novel in catalog or novel not in catalog). Right, predicted coding potential by CPAT. **j**, For novel isoforms with alternative TES, presence of poly(A) signals in the isoform and the matched GENCODE transcript was assessed via PAS motif and poly(A) predictions. Isoforms were further stratified by whether their 3' end overlaps a splice donor site. **k**, Venn diagram showing genes with both significantly up- and down-regulated isoforms across three comparisons. **l**, Venn diagram of genes with isoform switching involving alternative TSS, TES, and exon usage.